

Information Security : TP Fingerprinting

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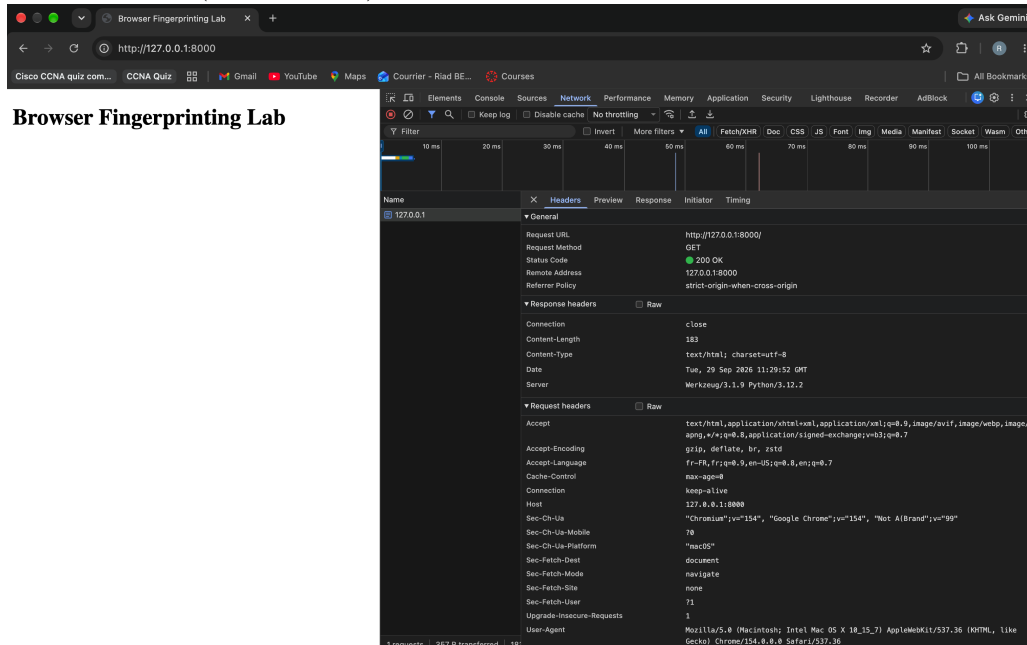
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1 Part 1

1.1 Task 0:

Before any JavaScript runs, the browser sends HTTP request headers to the server. We inspected them in DevTools (Network tab).



- **User-Agent:** browser, major version and OS family.
- **Accept-Language:** preferred languages and their order (**fr-FR**, **fr**, **en-US**, **en**). A rare combination is very identifying.
- **Accept:** supported content formats. The list differs between browsers.
- **Accept-Encoding:** supported compression (**gzip**, **deflate**, **br**, **zstd**). Separates recent browsers from old ones.
- **Sec-CH-UA:** browser brand and version, desktop or mobile, and OS.
- **Sec-Fetch:** request context (e.g. a navigation typed by the user).

1.2 Task 1:

We added five headers to `app.py`: **Accept**, **Accept-Encoding**, **Sec-CH-UA**, **Sec-CH-UA-Platform** and **Sec-Fetch-Dest**, then visited the page with Chrome and Brave.

```

Problems Output Debug Console Postman Console Terminal Ports
(venv) (base) → fingerprintingLab python app.py

127.0.0.1 -- [29/Sep/2026 12:06:56] "GET / HTTP/1.1" 200 -

===== NEW VISIT =====
IP address      : 127.0.0.1
HTTP method     : GET
User-Agent      : Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
Accept-Language : fr-FR,fr;q=0.9,en-US;q=0.8,en;q=0.7
Accept          : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding : gzip, deflate, br, zstd
Sec-CH-UA       : "Chromium";v="154", "Google Chrome";v="154", "Not A(Brand";v="99"
Sec-CH-UA-Platform : "macOS"
Sec-Fetch-Dest  : document

127.0.0.1 -- [29/Sep/2026 12:07:26] "GET / HTTP/1.1" 200 -

===== NEW VISIT =====
IP address      : 127.0.0.1
HTTP method     : GET
User-Agent      : Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
Accept-Language : en-US,en;q=0.9
Accept          : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Accept-Encoding : gzip, deflate, br, zstd
Sec-CH-UA       : "Chromium";v="154", "Brave";v="154", "Not A(Brand";v="99"
Sec-CH-UA-Platform : "macOS"
Sec-Fetch-Dest  : document

127.0.0.1 -- [29/Sep/2026 12:08:15] "GET / HTTP/1.1" 200 -
127.0.0.1 -- [29/Sep/2026 12:08:15] "GET /favicon.ico HTTP/1.1" 404 -

===== NEW VISIT =====
IP address      : 127.0.0.1
HTTP method     : GET
User-Agent      : Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
Accept-Language : en-US,en;q=0.9
Accept          : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Accept-Encoding : gzip, deflate, br, zstd
Sec-CH-UA       : "Chromium";v="154", "Brave";v="154", "Not A(Brand";v="99"
Sec-CH-UA-Platform : "macOS"
Sec-Fetch-Dest  : document

127.0.0.1 -- [29/Sep/2026 12:08:18] "GET / HTTP/1.1" 200 -

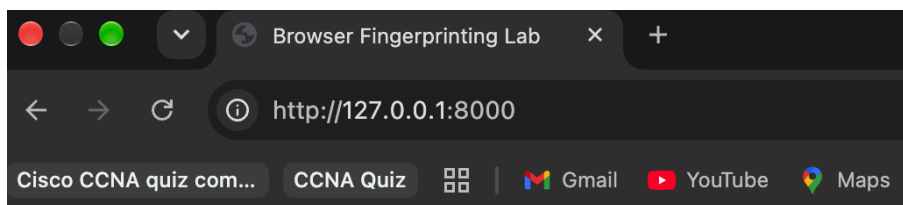
```

- **Same:** User-Agent, Accept-Encoding, Sec-CH-UA-Platform, Sec-Fetch-Dest. Brave uses exactly the Chrome User-Agent.
- **Different:** Sec-CH-UA contains "Brave" instead of "Google Chrome"; Accept-Language is shorter in Brave (en-US,en); Chrome's Accept adds application/signed-exchange.

So even with an identical User-Agent, the other headers still tell the two browsers apart.

2 Part 2

2.1 First Active Feature



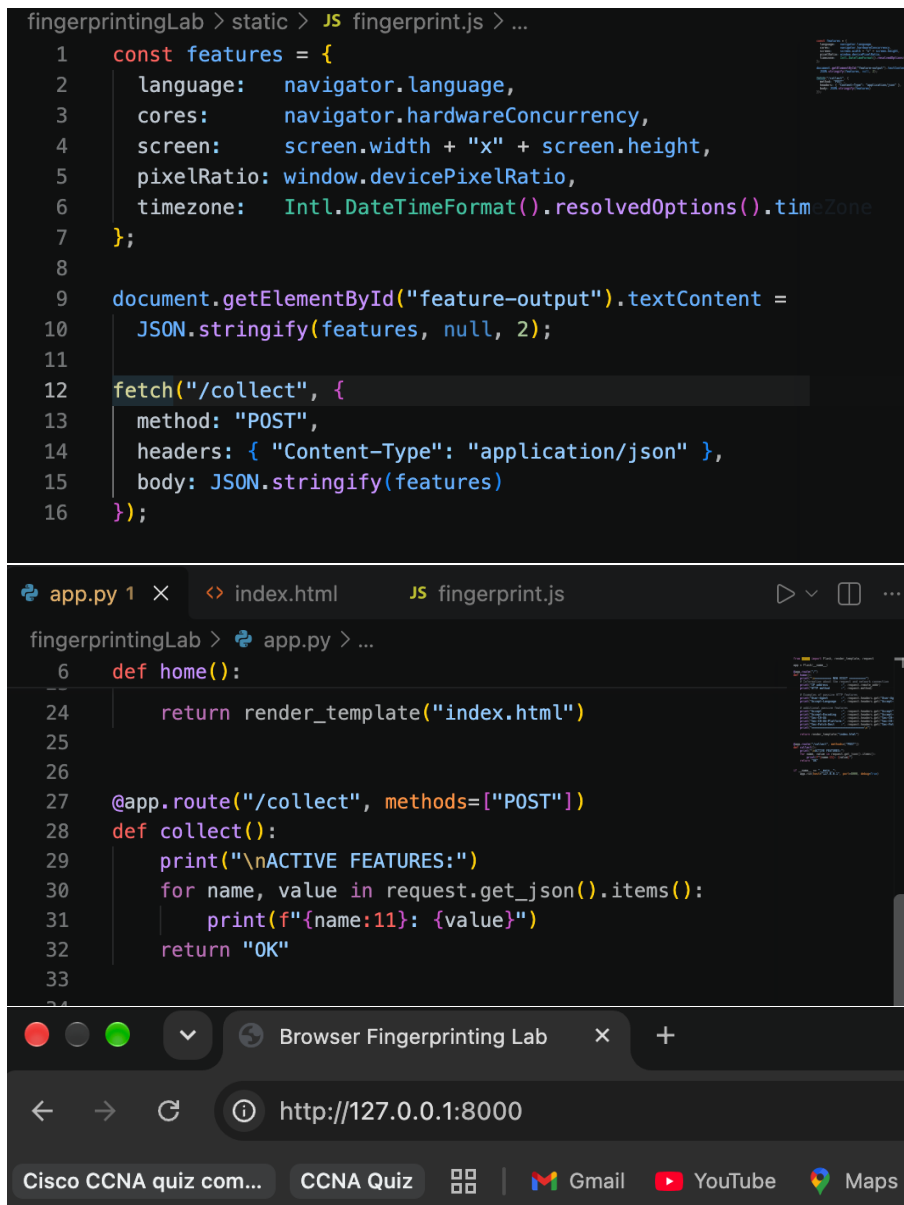
Browser Fingerprinting Lab

Active Feature Collection

Browser language: fr-FR

2.2 Task 2:

We updated `fingerprint.js` to collect four more features, display them on the page and send them to a new `/collect` route in Flask.



Browser Fingerprinting Lab

Active Feature Collection

```

{
  "language": "fr-FR",
  "cores": 8,
  "screen": "1470x956",
  "pixelRatio": 2,
  "timezone": "Africa/Casablanca"
}

```

```

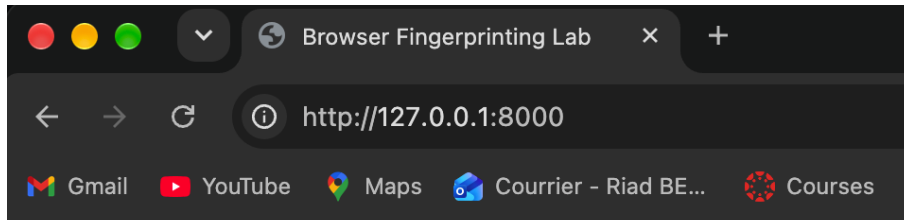
127.0.0.1 -- [29/Sep/2026 19:43:33] "GET / HTTP/1.1" 200 -
127.0.0.1 -- [29/Sep/2026 19:43:33] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
127.0.0.1 -- [29/Sep/2026 19:43:33] "GET /static/fingerprint.js HTTP/1.1" 200 -

ACTIVE FEATURES:
language : fr-FR
cores    : 8
screen   : 1470x956
pixelRatio : 2
timezone : Africa/Casablanca
127.0.0.1 -- [29/Sep/2026 19:43:33] "POST /collect HTTP/1.1" 200 -

```

- **CPU cores** (`navigator`): number of logical cores, differs between devices.
- **Screen** (`screen`): display resolution.
- **Pixel ratio** (`window`): screen density and zoom.
- **Time zone** (`Intl`): user's region.

Protection mechanisms:

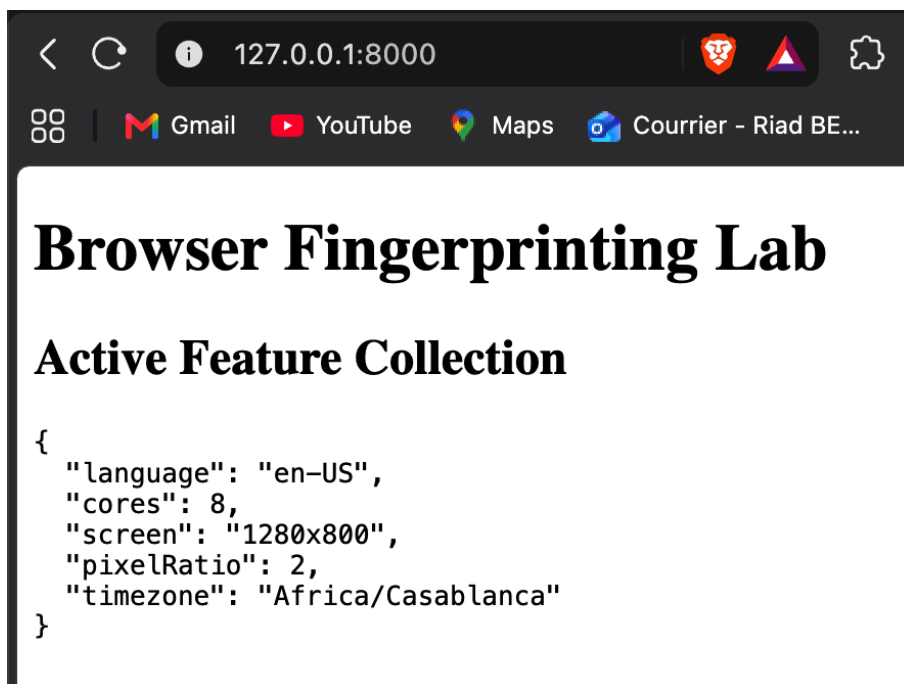


Browser Fingerprinting Lab

Active Feature Collection

```
{
  "language": "fr-FR",
  "cores": 8,
  "screen": "1470x956",
  "pixelRatio": 2,
  "timezone": "Africa/Casablanca"
}
```

Chrome Incognito



Brave

- **Chrome Incognito:** all features are identical to normal Chrome. Private mode only deletes cookies and history, and a fingerprint stores nothing in the browser.
- **Brave:** the screen resolution (1280x800 instead of 1470x956) and the language (**en-US**) change, while cores, pixel ratio and time zone stay the same.
- **Most effective:** Brave, because it changes some values and the fingerprint no longer matches Chrome's. **Limitation:** the remaining features: time zone, cores, pixel ratio.

3 Part 3

3.1 Task 3:

We added to `index.html` the target sentence, a text input, a restart button and an empty result area, each with a unique id.

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Browser Fingerprinting Lab</title>
6  </head>
7  <body>
8      <h1>Browser Fingerprinting Lab</h1>
9
10     <h2>Active Feature Collection</h2>
11     <pre id="feature-output">Waiting for JavaScript...</pre>
12
13     <script src="{{ url_for('static', filename='fingerprint.js') }}"></script>
14
15     <h2>Typing Test</h2>
16     <p id="target-sentence">Cookies are gone, but Alice's rhythm still leaves traces!</p>
17     <input id="typing-input" type="text" size="60">
18     <button id="restart-button">Restart</button>
19     <pre id="typing-results"></pre>
20
21     <script src="{{ url_for('static', filename='fingerprint2.js') }}"></script>
22 </body>
23 </html>

```

3.2 Task 4:

Typing speed:

$$\text{speed (chars/s)} = \frac{\text{number of characters in the sentence}}{\text{typing time (s)}}$$

Algorithm:

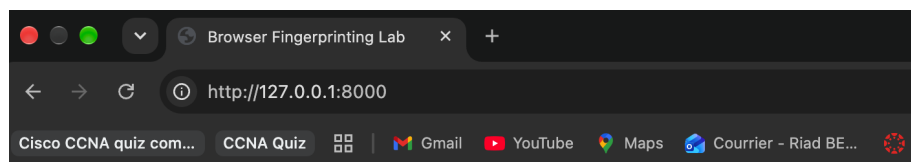
1. On the first character typed, save the start time.
2. On each Backspace, increase the correction counter.
3. After each change, compare the input with the sentence.
4. If they match, compute the time and speed, display them, and send them to `/collect`.
5. Restart resets the timer, the counter and the input.

4 Part 4

4.1 Optional Mission:

We combined the active features in a fixed order (`language|cores|screen|pixelRatio|timezone`), hashed the string with SHA-256 using `crypto.subtle.digest()`, displayed the result and sent only the hash to the server. The typing features were excluded because they change at every attempt.

```
app.py 1 index.html JS fingerprint.js X JS fingerprint2.js
fingerprintingLab > static > JS fingerprint.js > then() callback > hash
1  const features = {
4      screen:    screen.width + "x" + screen.height,
5      pixelRatio: window.devicePixelRatio,
6      timezone:   Intl.DateTimeFormat().resolvedOptions().timeZone
7  };
8
9  document.getElementById("feature-output").textContent =
10     JSON.stringify(features, null, 2);
11
12
13  const combined = [features.language, features.cores, features.screen,
14                   features.pixelRatio, features.timezone].join("|");
15
16
17  const data = new TextEncoder().encode(combined);
18  crypto.subtle.digest("SHA-256", data).then(function (buffer) {
19      const hash = Array.from(new Uint8Array(buffer))
20        .map(b => b.toString(16).padStart(2, "0")).join("");
21
22
23      document.getElementById("feature-output").textContent += "\n\nFingerprint: " + hash;
24      fetch("/collect", {
25        method: "POST",
26        headers: { "Content-Type": "application/json" },
27        body: JSON.stringify({ fp: hash })
28      });
29  });
```



Browser Fingerprinting Lab

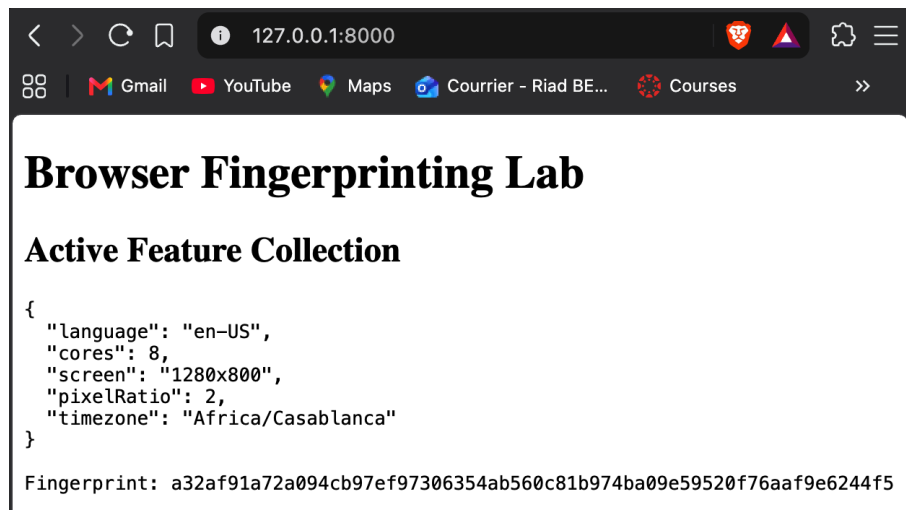
Active Feature Collection

```
{
  "language": "fr-FR",
  "cores": 8,
  "screen": "1470x956",
  "pixelRatio": 2,
  "timezone": "Africa/Casablanca"
}
```

Fingerprint: 01ee7a520a5afb9c5923e4d01118434aeba3e5e103e075f1d125389e21f7cbaf

```
127.0.0.1 - - [29/Sep/2026 21:29:57] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 21:29:57] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
127.0.0.1 - - [29/Sep/2026 21:29:57] "GET /static/fingerprint.js HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 21:29:57] "GET /static/fingerprint2.js HTTP/1.1" 304 -

ACTIVE FEATURES:
fp      : 01ee7a520a5afb9c5923e4d01118434aeba3e5e103e075f1d125389e21f7cbaf
127.0.0.1 - - [29/Sep/2026 21:29:57] "POST /collect HTTP/1.1" 200 -
█
```

- **Stability:** reloading the page in Chrome always gives the same hash (01ee7a52...7cbaf), so the server can recognize Alice without any cookie.
- **Chrome vs Brave:** only the language and the screen change, but the hash is completely different (a32af91a...244f5). With SHA-256, any small change gives an unrelated identifier.
- **Privacy:** the server only receives **fp**, so Alice sees a meaningless value and cannot tell what was collected.
- **Limitation:** if one feature changes (zoom, language...), the hash changes and the tracker loses Alice.